AKIB JAVED, Ph.D.

+880 1721 112202 akibjaved@outlook.com akibjaved
Google Scholar Scopus ResearchGate
ORCID @geoakib

Employment History

Apr 2026 – Present **Consultant on Hydrology and Geoinformatics, C4RE (Centre for Research on Environment and People)** Directs geospatial research and technical frameworks for evidence-based hydrological project roadmaps.

Mar 2026 – Present **Research Coordinator (Part-time), Center for People & Environment (CPE).** Coordinates the Research division by translating technical research into practical solutions and policy recommendations.

Feb 2026 – Present **Founder & Geospatial Consultant, Maplicon.** Established and leading a specialized consultancy firm providing research and consultancy services on remote sensing analysis, Google Earth Engine (GEE) automation, and spatial data science solutions.

Feb 2025 – Jan 2026 **GIS Analyst, Institute of Water Modelling (IWM).** Georeferenced and digitized old cadastral maps for a national land zoning project, improving land management efficiency for government agencies and reducing manual processing time by 20% within my team.

Jun 2025 – Aug 2025 **GIS Supervisor (Part-time), Center for People & Environment (CPE).** Led a GIS team of 3 analysts, improving project delivery speed by 20% and ensuring 95% accuracy in all spatial data outputs.

Jun 2019 – Jun 2019 **Internship, Hi-Target Surveying Instrument Co., Ltd., Guangzhou, China.** Participated in a week-long internship program and familiarized with the latest geospatial tools and software.

Jul 2017 – Jul 2017 **Visiting Lecturer.** Taught GIS and remote sensing courses to 100+ undergraduate students, improving average student geospatial knowledge by 5% through hands-on sessions within a week.

Apr 2017 – Aug 2017 **Research Assistant.** Analyzed land use patterns for the Adivasi People project, generating key insights that informed community development strategies for 3,000+ beneficiaries.

Education

Academic Degrees

2017 – 2024 **Ph.D.** in **Cartography and GIS** from LIESMARS, Wuhan University, China.

2014 – 2015 **M.S.** in **Geography and Environmental Studies** from University of Rajshahi.

2010 – 2013 **B.Sc.** in **Geography and Environmental Studies** from University of Rajshahi.

Scholarship

2017 **CSC Scholarship**: Awarded by Chinese Scholarship Council for doctoral studies

Research

Academic Service

Journal Peer Review	Remote Sensing of Environment, Advances in Space Research, Journal of Selected Topics in Applied Earth Observations and Remote Sensing, City and Environment Interactions
Conference Review	ISPRS Geospatial Week 2025, Dubai

Research Co-supervision

2025	Spatio-temporal Mapping of Forest Areas of Bangladesh: A remote sensing base study using Google Earth Engine (GEE) platform.
2024	Spatiotemporal Analysis of Droughts in Northern Bangladesh using MODIS Time Series (2000-2023).
	Spatiotemporal Mapping of Urban Air Pollutants: A Comparative Study of Bangladesh.

Research Experience

PhD thesis	Researched urban impervious surface extraction across space and time using open-source spatial big data.
B.Sc. project	Investigated urban transformation in Rajshahi City, revealing key patterns of urban sprawl driven by road networks and informing sustainable development strategies for 500K+ residents.
2017	Conducted intensive land study for the Adivasi People project, collecting data from 300+ households and informing community development strategies for 5K+ beneficiaries.
2013	Authored a detailed field reports from a socioeconomic survey on Ashariadaha Island, providing critical data that informed local lifestyle and economic activities for 3K+ residents.
2012	Authored a comprehensive field reports from a socioeconomic survey on Dubla Char Island, delivering key insights of local fisherman.
2011	Authored a field reports on industrial regions of Kushtia, providing critical data that informed major economic activities for four factories.

Research Publications

Journal Articles

1. M. N. Ahmad, X. Yuan, L. Gu, S. Yang, R. W. Aslam, **A. Javed**, and G. D. Bathrellos, "Advancements in deep learning methods for precise weed and rice classification from UAV imagery: A comprehensive review," *Discover Agriculture*, vol. 4, no. 1, p. 8, Jan. 2026, ISSN: 2731-9598. DOI: 10.1007/s44279-025-00469-0

2. M. N. Ahmad, X. Yuan, S. Yang, L. Gu, H. Peng, R. W. Aslam, M. E. Huq, **A. Javed**, and G. D. Bathrellos, "Dtsm: A uav-based framework for temporal rice stress detection and growth analysis," *Smart Agricultural Technology*, p. 102 141, 2026, ISSN: 2772-3755. DOI: https://doi.org/10.1016/j.atech.2026.102141

3. I. Ara, R. I. Zerin, L. A. Eva, M. K. Hossain, M. H. Mou, **A. Javed**, S. Ali, M. A. Kabir, and M. M. P. Rana, "Comparative hydrogeochemical characterization and environmental controls on groundwater quality, irrigation suitability, and health risk across northern and coastal hydrological settings of bangladesh," *Cleaner Water*, vol. 5, p. 100 194, 2026, ISSN: 2950-2632. DOI: https://doi.org/10.1016/j.clwat.2025.100194

4. M. N. Ahmad, F. Almutlaq, M. E. Huq, F. Islam, **A. Javed**, H. D. Skilodimou, and G. D. Bathrellos, "Enhanced urban impervious surface land use mapping using a novel multi-sensor feature fusion method and remote sensing data," *Environmental Earth Sciences*, vol. 84, no. 9, p. 228, Apr. 2025, ISSN: 1866-6299. DOI: 10.1007/s12665-025-12217-0

5. M. N. Ahmad, H. D. Skilodimou, F. Islam, **A. Javed**, and G. D. Bathrellos, "Drainage network generation for urban pluvial flooding (upf) using generative adversarial networks (gans) and gis data," *Sustainability*, vol. 17, no. 10, 2025, ISSN: 2071-1050. DOI: 10.3390/su17104380

6. R. I. Zerin, I. Ara, S. Ali, M. K. Hossain, **A. Javed**, S. N. Ratna, and M. A. Kabir, "Geostatistical modeling of soil physicochemical properties and environmental drivers in agro-industrial landuse systems in bangladesh," *Geomatica*, vol. 77, no. 2, p. 100 075, Sep. 2025, ISSN: 1195-1036. DOI: https://doi.org/10.1016/j.geomat.2025.100075

7. M. N. Ahmad, Z. Shao, **A. Javed**, I. Ahmad, F. Islam, H. D. Skilodimou, and G. D. Bathrellos, "Optical-sar data fusion based on simple layer stacking and the xgboost algorithm to extract urban impervious surfaces in global alpha cities," *Remote Sensing*, vol. 16, no. 5, 2024. DOI: 10.3390/rs16050873

8. M. N. Ahmad, Z. Shao, X. Xiao, P. Fu, **A. Javed**, and I. Ara, "A novel ensemble learning approach to extract urban impervious surface based on machine learning algorithms using sar and optical data," *International Journal of Applied Earth Observation and Geoinformation*, vol. 132, p. 104 013, 2024, ISSN: 1569-8432. DOI: https://doi.org/10.1016/j.jag.2024.104013

9. M. E. Huq, X. Wu, **A. Javed**, Y. Dong, B. Li, D. Lopez-Carr, J. Wu, J. Liu, Y. Zhang, F. Zhang, M. Riaz, M. N. I. Sarker, W. Soufan, K. F. Almutairi, and A. Tariq, "Changes monitoring in hongjiannao lake from 1987 to 2023 using google earth engine and analysis of climatic and anthropogenic forces," *Physics and Chemistry of the Earth, Parts A/B/C*, vol. 136, p. 103 756, 2024. DOI: https://doi.org/10.1016/j.pce.2024.103756

10. **A. Javed**, Z. Shao, I. Ara, M. N. Ahmad, E. Huq, N. Saleem, and F. Karim, "Development of soil-suppressed impervious surface area index for automatic urban mapping," *Photogrammetric Engineering & Remote Sensing*, vol. 90, no. 1, pp. 33–43, 2024, ISSN: 0099-1112. DOI: 10.14358/PERS.23-00043R2

11. Z. Shao, M. N. Ahmad, and **A. Javed**, "Comparison of random forest and xgboost classifiers using integrated optical and sar features for mapping urban impervious surface," *Remote Sensing*, vol. 16, no. 4, p. 665, 2024, ISSN: 2072-4292. URL: https://www.mdpi.com/2072-4292/16/4/665

12. M. N. Ahmad, Z. Shao, and **A. Javed**, "Modelling land use/land cover (lulc) change dynamics, future prospects, and its environmental impacts based on geospatial data models and remote sensing data," *Environmental Science and Pollution Research*, vol. 30, no. 12, pp. 32 985–33 001, 2023, ISSN: 1614-7499 (Electronic) 0944-1344 (Linking). DOI: 10.1007/s11356-022-24442-2

13. M. N. Ahmad, Z. Shao, and **A. Javed**, "Mapping impervious surface area increase and urban pluvial flooding using sentinel application platform (snap) and remote sensing data," *Environmental Science and Pollution Research*, vol. 30, no. 60, pp. 125 741–125 758, 2023, ISSN: 1614-7499. DOI: 10.1007/s11356-023-30990-y

14. M. N. Ahmad, Z. Shao, **A. Javed**, F. Islam, H. H. Ahmad, and R. W. Aslam, "The cellular automata approach in dynamic modelling of land use change detection and future simulations based on remote sensing data in lahore pakistan," *Photogrammetric Engineering & Remote Sensing*, vol. 89, no. 1, pp. 47–55, 2023, ISSN: 0099-1112. DOI: 10.14358/pers.22-00102r2

15. M. Y. Ali, M. Shahrier, A. A. Kafy, I. Ara, **A. Javed**, M. A. Fattah, Z. A. Rahaman, and K. Tripura, "Environmental impact assessment of tobacco farming in northern bangladesh," *Heliyon*, vol. 9, no. 3, e14505, 2023, ISSN: 2405-8440 (Print) 2405-8440 (Electronic) 2405-8440 (Linking). DOI: 10.1016/j.heliyon.2023.e14505

16. **A. Javed**, Z. F. Shao, B. Bai, Z. Y. Yu, J. B. Wang, I. Ara, M. E. Huq, M. Y. Ali, N. Saleem, M. N. Ahmad, N. S. Sumari, and Mardia, "Development of normalized soil area index for urban studies using remote sensing data," *Geofizika*, vol. 40, no. 1, pp. 29–49, 2023, ISSN: 0352-3659. DOI: 10.15233/gfz.2023.40.2

17. Z. Shao, M. N. Ahmad, **A. Javed**, F. Islam, Z. Jahangir, and I. Ahmad, "Expansion of urban impervious surfaces in lahore (1993-2022) based on gee and remote sensing data," *Photogrammetric Engineering and Remote Sensing*, vol. 89, no. 8, pp. 479–486, 2023, ISSN: 0099-1112. DOI: 10.14358/pers.23-00001r2

18. M. N. Ahmad, Z. F. Shao, A. Yaseen, M. N. Khalid, and **A. Javed**, "The simulation and prediction of land surface temperature based on scp and ca-ann models using remote sensing data: A case study of lahore,"

Photogrammetric Engineering and Remote Sensing, vol. 88, no. 12, pp. 783–790, 2022, ISSN: 0099-1112. DOI: 10.14358/pers.22-00071r2

19. M. Y. Ali, A.-A. Kafy, Z. A. Rahaman, M. F. Islam, M. R. Rahman, I. Ara, M. R. Akhtar, and **A. Javed**, "Comparative occupational health risk between tobacco and paddy farming people in bangladesh," *SSM-Mental Health*, vol. 2, p. 6, 2022.

20. **A. Javed**, Q. M. Cheng, H. Peng, O. Altan, Y. Li, I. Ara, E. Huq, Y. Ali, and N. Saleem, "Review of spectral indices for urban remote sensing," *Photogrammetric Engineering and Remote Sensing*, vol. 87, no. 7, pp. 513–524, 2021, ISSN: 0099-1112. DOI: 10.14358/Pers.87.7.513

21. M. E. Huq, Q. M. Cheng, O. Altan, A. Z. M. Shoeb, M. A. Hossain, M. N. I. Sarker, N. Saleem, **A. Javed**, X. Long, A. A. Al Dughairi, M. M. P. Rana, A. Al Mamun, and M. M. Rahman, "Assessing vulnerability for inhabitants of dhaka city considering flood-hazard exposure," *Geofizika*, vol. 37, no. 2, pp. 97–130, 2020. DOI: 10.15233/gfz.2020.37.5 WOS: 000608071700001. Accessed: Oct. 7, 2025.

22. N. Saleem, M. E. Huq, N. Y. D. Twumasi, **A. Javed**, and A. Sajjad, "Parameters derived from and/or used with digital elevation models (dems) for landslide susceptibility mapping and landslide risk assessment: A review," *ISPRS International Journal of Geo-Information*, vol. 8, no. 12, pp. 545–572, 2019.

Books and Chapters

1. M. E. Huq, D. Lopez-Carr, A. Tariq, X. Wu, J. Wu, Y. Dong, Z. Daud, **A. Javed**, M. N. I. Sarker, K. M. Adnan, et al., "Advancements in remote sensing and gis for water resource management and sdg monitoring," in *Utilizing Earth Observation Data in Reaching Sustainable Development Goals*. Elsevier, 2026, pp. 189–218. DOI: https://doi.org/10.1016/B978-0-443-30204-6.00007-3

2. M. E. Huq, Z. Shao, A. A. Al Dughairi, M. N. I. Sarker, C. Bowen, A. Al Mamun, N. Saleem, **A. Javed**, and M. M. Rahman, "Measuring vulnerability to flash flood of urban dwellers," in *Wadi Flash Floods*. Springer, Singapore, 2022, p. 317.

3. M. E. Huq, A. Z. M. Shoeb, M. A. Hossain, S. Fahad, M. M. Kamruzzaman, **A. Javed**, N. Saleem, K. M. M. Adnan, S. A. Sarker, and M. Y. Ali, "Measuring vulnerability to environmental hazards: Qualitative to quantitative," in *Environment, Climate, Plant and Vegetation Growth*. Springer, 2020, pp. 421–452.

Conference Proceedings

1. M. S. Akter, I. Ara, and **A. Javed**, "A comprehensive evaluation of environmental literacy: A case study in kushtia municipality, bangladesh," in *ABSTRACTS BOOK OF FIRST INTERNATIONAL CONFERENCE ON BANGLADESH STUDIES (ICBS-I) 2025*, O. AKGÜL, H. Rahman, and A. M. Hasan, Eds., Presented via online, Istanbul Ticaret University, Türkiye: Istanbul Ticaret University, Türkiye, Apr. 2025, p. 24.

2. M. S. Akter, I. Ara, and **A. Javed**, "A comprehensive evaluation of environmental literacy: A case study in kushtia municipality, bangladesh," in *Conference Proceedings on 1st International Conference on Bangladesh Studies (ICBS-I), 2025*, Hybrid conference held on April 19-20, 2025, Istanbul Ticaret University, Türkiye, Apr. 2025. URL: https://cpsrinfo.org/

3. S. Ali, **A. Javed**, and I. Ara, "Agricultural drought in northern bangladesh: A spatiotemporal analysis using modis time series from 2001 to 2023," in *ABSTRACTS BOOK OF FIRST INTERNATIONAL CONFERENCE ON BANGLADESH STUDIES (ICBS-I) 2025*, O. AKGÜL, H. Rahman, and A. M. Hasan, Eds., Presented via online, Istanbul Ticaret University, Türkiye: Istanbul Ticaret University, Türkiye, Apr. 2025, p. 46.

4. L. A. Eva, I. Ara, M. S. K. Khan, M. A. Rahman, and **A. Javed**, "From resource to risk: Evaluating the rainwater quality and associated health issues in the southern coastal bangladesh," in *Conference Proceedings on 10th International Conference on Water and Flood Management - ICWFM 2025*, vol. 10, Institute of Water and Flood Management, BUET, Feb. 2025, pp. 277–284.

5. F. Islam, I. Ara, M. S. Khan, and **A. Javed**, "Assessment of particulate matter exposure and associated human health risk: Comparison among non-automated, automated and semi-automated rice processing industry workers in bangladesh," in *Proceedings of the 1st International Conference on Geosciences for Achieving Sustainable Development Goals (ICGSDG-2025)*, Faculty of Geosciences, University of Rajshahi, Bangladesh, Nov. 2025, p. 162.

6. H. Rashid, **A. Javed**, and I. Ara, "A Remote Sensing Assessment of Air Pollution in Five Bangladeshi Industrial Cities," in *INTERNATIONAL GLOBAL SUSTAINABILITY AND DEVELOPMENT CONGRESS Full Text Book*, O. Özkan and Ş. T. Azgin, Eds., New York, NY: Liberty Publishing House, Nov. 2025, pp. 873–897, ISBN: 979-8-89695-250-3. URL: iksadkongre.com/_files/ugd/614b1f_c112d6b2a09442749bd2d5579d9cdb53.pdf

7. H. Rashid, **A. Javed**, and I. Ara, "A remote sensing assessment of air pollution in five bangladeshi industrial cities," in *Global Sustainability and Development Congress Abstract Book*, O. Özkan and Ş. T. Azgın, Eds., Kayseri, Türkiye: Liberty Academic Publishers, Oct. 2025, p. 245, ISBN: 979-8-89695-212-1.

8. R. R. Ria, I. Ara, and **A. Javed**, "An analysis of governance framework effectiveness and internal waste management practices in the hospitals of rangpur sadar upazila, bangladesh," in *ABSTRACTS BOOK OF FIRST INTERNATIONAL CONFERENCE ON BANGLADESH STUDIES (ICBS-I) 2025*, O. AKGÜL, H. Rahman, and A. M. Hasan, Eds., Presented via online, Istanbul Ticaret University, Türkiye: Istanbul Ticaret University, Türkiye, Apr. 2025, p. 77.

9. R. R. Ria, I. Ara, and **A. Javed**, "An analysis of governance framework effectiveness and internal waste management practices in the hospitals of rangpur sadar upazila, bangladesh," in *Conference Proceedings on 1st International Conference on Bangladesh Studies (ICBS-I), 2025*, Hybrid conference held on April 19-20, 2025, Istanbul Ticaret University, Türkiye, Apr. 2025. URL: https://cpsrinfo.org/

10. R. I. Zerin, I. Ara, L. Anjum, M. H. Moti, M. K. Hossain, and **A. Javed**, "Index-based assessment of groundwater quality and suitability for irrigation across major agricultural regions of bangladesh," in *Proceedings of the 1st International Conference on Geosciences for Achieving Sustainable Development Goals (ICGSDG-2025)*, Faculty of Geosciences, University of Rajshahi, Bangladesh, Nov. 2025, p. 109.

11. M. Huq, S. Zhenfeng, C. Bowen, N. Saleem, A. A. Aldughairi, D. T. N. Yaw, **A. Javed**, and M. M. Rahman, "Measuring vulnerability to flash flood for urban inhabitants: A case study of dhaka megacity, bangladesh," in *The Fifth International Symposium on Flash Floods in Wadi Systems (ISFF2020)*, 2020, p. 87.

12. M. E. Huq, A. Shoeb, **A. Javed**, Z. Shao, M. A. Hossain, and M. S. Sarven, "Measuring vulnerability for city dwellers exposed to flood hazard: A case study of dhaka city, bangladesh," in *Urban Intelligence and Applications: Proceedings of ICUIA 2019*, Springer International Publishing, 2019, pp. 207–215, ISBN: 3030450988.

13. M. Y. Ali, M. Hassan, M. A. Rahmana, A.-A. Kafy, I. Ara, **A. Javed**, and M. R. Rahman, "Life cycle energy and cost analysis of small scale biogas plant and solar pv system in rural areas of bangladesh," in *2nd International Conference on Energy and Power, ICEP2018, 13–15 December 2018, Sydney, Australia*, vol. 160, Energy Procedia, ELSEVIER, 2018, pp. 277–284.

14. **A. Javed** and S. Zhenfeng, "Smart application using gnss, gis and remote sensing: Flood prediction and monitoring," in *First International Workshop on Artificial Intelligence and Big Data Analysis 2017*, 2017, p. 67.

15. **A. Javed** and R. Ahmed, "Philosophy of geography education: Some perspectives and progress in bangladesh context," in *International Conference on Geography Education 2015*, 2015, p. 33.

Posters

1. **A. Javed** and M. N. Ahmad, "Chinese desert reclamation: Turning barren land into green oases," Mar. 2023. DOI: 10.13140/RG.2.2.30223.46240

2. **A. Javed**, Z. Shao, I. Ara, M. N. Ahmad, E. Huq, N. Saleem, and F. Karim, "Soil suppressed generalized impervious surface area index (gisai) for automatic megacity mapping," Mar. 2023. DOI: 10.13140/RG.2.2.35676.05761

Seminars, Conferences, Workshops, Trainings

Seminars

2023-10-12 **Mobile Mapping Systems: Demonstrated Technology that Keeps Evolving** by Prof. Ayman F. Habib, at LIESMARS, Wuhan University.

2023-10-03 **Unleashing the Power of Words: Writing Practices and Processes** by Stephen C. McClure, at LIESMARS, Wuhan University.

2023-06-20 **Smart City** by Prof. Dr. Zhenfeng Shao, at LIESMARS, Wuhan University.

2023-05-30 **Navigating the Academic Landscape: Insights and Strategies for Successful Research and Career Development** by Prof. Naser El-Sheimy, at LIESMARS, Wuhan University.

2023-05-21 **Forest Aboveground Biomass Estimation with Airborne LiDAR data** by Prof. Dengsheng Lu, at LIESMARS, Wuhan University.

2023-05-06 **LEO Navigation Augmentation System from Luojia-1A/2A Satellite - Design, Progress and Application** by Prof. Lei Wang, at LIESMARS, Wuhan University.

Conferences

17-19 November 2024 Participated in the **2024 International Symposium on Spatio-Temporal Intelligence and 2024 International Conference on Geospatial Information Science: Education, Innovation and Applications** organized by LIESMARS, Wuhan University.

15-16 October 2023 Participated in the **International Conference on Geospatial Information Science: Education, Innovation and Applications** organized by LIESMARS, Wuhan University.

16-18 October 2018 Participated in the **International Conference on Advance Remote Sensing 2018** organized by LIESMARS, Wuhan University.

20 December 2017 Participated in the **International Symposium on key technologies and applications of Video Surveillance and Intelligent Analysis in Hengqin Island** organized by Zhuhai City Development Authority, Guangzhou, Guangdong, China.

11-13 September 2015 Participated in the **International Conference on Geography Education 2015** organized by the Dept. of Geography and Environmental Studies, University of Rajshahi, Bangladesh and Institute of Geography, University of Freiburg, Germany at the University of Rajshahi, Bangladesh.

07-08 March 2014 Participated in the **International Conference on Environment and Development** organized by the Department of Geography and Environmental Studies, University of Rajshahi, supported by Higher Education Quality Enhancement Project (HEQEP).

Workshop

2019 Ten days long Workshop title **Zhuhai One Hyperspectral Remote Sensing Training** organized by Zhuhai ORBITA Aerospace Science & technology Co., Ltd. at Obit Science and Technology Park, No. 1 Baisha Road, Tangjia East Bank, Zhuhai City, China.

2017 Three days long workshop title **The 1st LIESMARS International Graduate Workshop on GeoInformatics** organized by LIESMARS, Wuhan University, Wuhan City, China (November 25-27, 2017).

2014 Participated to a Workshop on **Teaching Method and Curriculum Development** at the Department of Geography and Environmental Studies, University of Rajshahi, Bangladesh: Organized by University of Education Freiburg, Germany and University of Rajshahi, Bangladesh (November 10-12, 2014).

Seminars, Conferences, Workshops, Trainings (continued)

2013 — Participated on Teacher Training Workshop on **Innovative Teaching Methodology, Learning Strategies and Didactics** at the University of Rajshahi, Bangladesh. Organized by the University of Education Freiburg, Germany and the University of Rajshahi, Bangladesh (18-20 February 2013).

Training and Summer School

2024 — Completed ten-day long training Program titled **Application of Mathematical Modelling & GIS Technology for Feasibility Study of Water Development Projects** organized by Institute of Water Modelling (IWM), Dhaka, Bangladesh (29 September - 09 October 2024).

23-29 June, 2024 — Participated in **2024 Luojia Global Summer School** organized by Wuhan University. Took courses on **Social Sensing and Big Data Computing for Disaster Management** with the instructor Prof Dr. Zhenlong Li from The Pennsylvania State University, USA.

12-18 July, 2023 — Participated in **2023 International Geo-Informatics Summer School** organized by LIESMARS, Wuhan University. Took courses on **Computer Vision and Digital Image Processing** with the instructor Dr. Peng Fu from Harrisburg University, USA.

01-07 July, 2019 — Participated in **2019 Geo-Informatics Summer School** organized by LIESMARS, Wuhan University and Collaborative Innovation Center of Geospatial Technology. Took courses on **Microwave Remote Sensing** with the instructor Prof. Dr. Timo Balz.

2014 — Completed three-month long training program titled **Statistical Computing Certificate Course on MS Excel, SPSS and Programming with R** organized by Department of Statistics, University of Rajshahi, Bangladesh.

Certifications

2026
- **Advanced Prompt Engineering Techniques** (LinkedIn)
- **How to Be an Agent Boss: Leading AI Agents at Work** (LinkedIn)
- **How to Learn a New Topic Quickly** (LinkedIn)
- **Power BI: Dashboards for Beginners** (LinkedIn)
- **Foundations of Geographic Information Systems (GIS)** (LinkedIn)

2025
- **Microsoft Copilot: The Art of Prompt Writing** (LinkedIn)
- **Introduction to Career Skills in Software Development** (LinkedIn)
- **Programming Foundations: Beyond the Fundamentals** (LinkedIn)
- **Programming Foundations: Fundamentals** (LinkedIn)
- **Practical Project Management: Gantt Charts, PMI, PMP, APM** (Udemy)

2024
- **AI Python for Beginners: Basics of AI Python Coding** (DeepLearning.AI)

2022
- **Introduction to Python** (DataCamp)
- **Intermediate Python** (DataCamp)
- **Data Science for Everyone** (DataCamp)
- **Learning Excel: Data Analysis** (LinkedIn)
- **Learning Data Analytics: 1 Foundations** (LinkedIn)
- **Communication Foundations** (LinkedIn)
- **Writing in the Sciences** (Coursera)
- **Good Citation Behavior** (Clarivate)

Certifications (continued)

- **Reviewing in the Sciences** (Clarivate)

Technical Skills

Programming & DB	Python, JavaScript, R, SQL, LaTeX
GIS & Cloud	Google Earth Engine (GEE), ArcGIS Desktop, QGIS
AI & ML Models	Machine Learning (Random Forest, XGBoost), Deep Learning Architectures, Neural Networks, Ensemble Learning, Cellular Automata, Advanced Prompt Engineering
Data & Analytics	Power BI, IBM SPSS Statistics, MS Excel, Google Sheets
Field Surveying	RTK, GPS, Total Station, UAVs (Data Acquisition & Processing)

Research Expertise

Remote Sensing & GIS	•Generalized urban impervious surface area mapping, urban land cover classification, dynamic LULC change detection, and multi-sensor data fusion (Simple Layer Stacking). •Image processing and texture analysis (GLCM) utilizing optical, SAR, and AlphaEarth's Satellite Embeddings (64 bands).
Spatial Modeling	•Development of novel spectral indices (GISAI, NSAI); predictive modeling, environmental hazard assessment, and risk mapping (e.g., flood prediction).
Research Methodology	•Experimental design, quantitative & qualitative analytics, sampling (10-meter resolution scale), socio-economic surveys, academic citations, and peer review.
Management & Strategy	•Project Management (PMP, APM), Gantt Charts, interdisciplinary collaboration, team leadership, public speaking, and SEO.

Language Proficiency

Bengali	Native speaker.
English	Professional Command. IELTS test score: overall 6.5.
Dialects	Comprehension of Asian English dialects including Indian English and Chinese English.
Urdu	Conversational proficiency.
Hindi	Conversational proficiency.
Chinese	HSK 1. Taking online course for HSK 3 exam.
Arabic	Can read. Taking online course for improve overall comprehension.

Extracurricular Activities

EGSC

- **English GeoScience Cafe (EGSC)**, LIESMARS, Wuhan University
  - **Editorial Member** (2023–2024): Managed academic content
  - **Vice Director** (2017–2018): Organized seminars/conferences
  - Coordinated field trips and volunteer programs

SERC

- **Space & Environment Research Center** (Co-founder)
  - **General Secretary** (2014–2015)
  - Organized national conference on disaster management

Extracurricular Activities (continued)

RUPC **Rajshahi University Press Club**
- **Elected Treasurer** (2012–2014)
- **Senior Executive Member** (2014–2015)
- **Rajshahi University Correspondent** of Daily Janata (2011–2015)

BNCC **Bangladesh National Cadet Corps (BNCC)**
- **Cadet Corporal** (2010–2013): Rajshahi University Naval Branch, Khulna Flotilla
- **Contingent Commander** (2013): Led the 'Freedom Day Parade' at the University of Rajshahi
- Completed a **12-day Annual Training Camp** at BNS Shaheed Moazzam, Rangamati
- Completed a **5-day Capsule Training Camp** at Khulna Naval Flotilla, Khulna

RUAA **Rajshahi University Alumni Association**
- Life Member (2025–Present)

References

Dr. Md. Enamul Huq
Senior Lecturer
Universiti Pendidikan Sultan Idris (UPSI), Tanjung Malim, Perak, Malaysia
mdenamulhuq@fsk.upsi.edu.my
polton86@gmail.com
+60 11-4112 0664 (Whatsapp)

Dr. Md Masud Parves Rana
Professor
Department of Geography and Environmental Studies
University of Rajshahi, Bangladesh
mprges@ru.ac.bd
mmprana@yahoo.com
+880-1712-127-621 (Whatsapp)

Dr. Md. Moniruzzaman
Professor
Department of Geography and Environmental Studies
University of Rajshahi, Bangladesh
mzges@ru.ac.bd
moniralvi@gmail.com
+880-1796-584-574 (Whatsapp)

Muhammad Abdur Rahaman
Director
Center for People & Environ (CPE)
Lalmatia, Dhaka-1207
info@cpe-bd.org
+880-1973-487838 (Whatsapp)